Self-Confidence in Math and STEM Major Choice at CUHK-Shenzhen

URC2026 Presentation

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Agenda

1 Introduction & Literature

2 Research Questions

3 Methodology

4 Key Findings

5 Impact & Conclusion

Introduction and Motivation

The Key Insight

Choosing a STEM major is not just about academic ability; it is fundamentally about **believing** in that ability.

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The Key Insight

Choosing a STEM major is not just about academic ability; it is fundamentally about **believing** in that ability.

- **Why this is important:**
  - Math confidence influences STEM major choices as much as academic performance (Escobar et al., 2023).
  - If students lack confidence, they may filter themselves out of STEM fields even if they are capable (Wang, 2013).
  - Understanding this helps educators support students in making choices based on *genuine interest* rather than fear.

Literature Review & The Research Gap

What is known?

- **Prediction:** Math confidence strongly predicts STEM pursuit (Moakler Jr and Kim, 2014).
- **Disconnect:** Confidence $\neq$ Performance (Ferdinand et al., 2024).
- **East Asian Paradox:** Students in East Asia often perform highly but report low confidence (Leung, 2017).

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The Gap: Sino-foreign universities

Does this paradox persist in **CUHK-Shenzhen**:

- High internationalization ($\sim$15%).
- **Flexible major switching**.

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This unique context allows me to test if confidence drives choice when students have the freedom to act on it.

Research Aim and Questions

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Research Questions (All in CUHK-SZ setting):

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2. Does math confidence predict actual math performance?
3. Are Chinese students more confident than international students?
4. What external factors affect math confidence?

Methodology

Research Design:

- Quantitative Online Survey ($N = 91$).
- **Ethics:** Anonymous, voluntary participation
- 82.4% Chinese, 17.6% International.
- 61.5% STEM, 38.5% Non-STEM.

Key Measures:

1. **Math Confidence:** 9-item Likert scale.
2. **Performance:** Weighted self-reported scores (eg. 60% University & 40% High School).
3. **Influencing Factors:** Multi-select.

Data Analysis: Independent T-tests & Pearson Correlation Analysis.

RQ1: Confidence and Major Choice

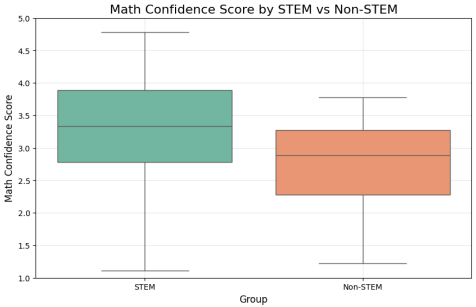


Figure: Confidence by Major

Result: Significant Difference

STEM students ($M = 3.26$) reported significantly higher confidence than Non-STEM students ($M = 2.76$) with $p = .003$.

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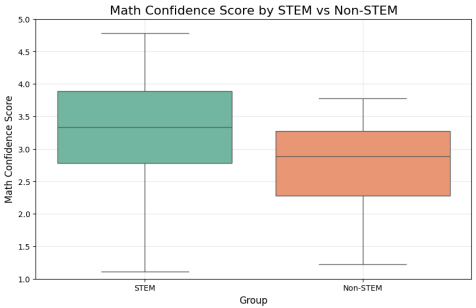


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Interpretation

Confidence acts as a **filter**. In CUHK-SZ, students with higher self-belief in math are more likely to choose STEM, while others may avoid doing so.

RQ2: Confidence vs. Performance

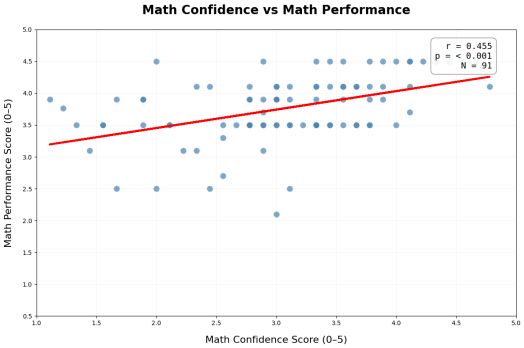


Figure: Correlation Analysis

Result:

Moderate positive correlation ($r = 0.455, p < .001$).

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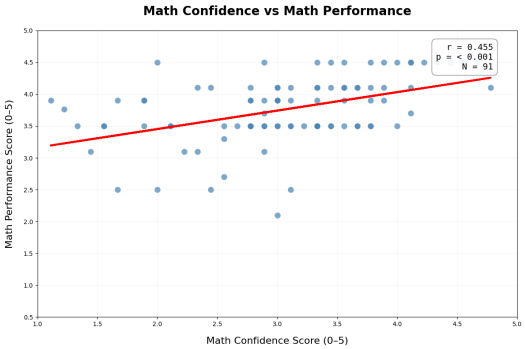


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Interpretation:
While confidence and performance are linked, they are **distinct constructs**. This suggests that confidence is a separate psychological lever we can target to improve academic outcomes.

RQ3: The "Culture" Surprise (Result)

Hypothesis vs. Reality

Literature suggests Chinese students would report lower confidence due to cultural humility/highly competitive environment.

Actual Result: No significant difference ($p = .705$).

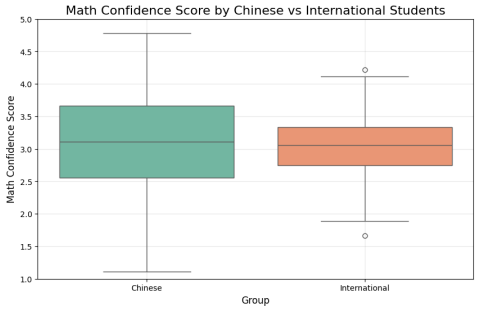


Figure: Confidence by Nationality

RQ3: The "Culture" Surprise (Interpretation)

Why No Difference?

Interpretation:

- The "Micro-environment" at CUHK-SZ (intense workload, peer pressure) outweighs broad cultural norms.
- International students likely assimilate into the local high-pressure academic culture.

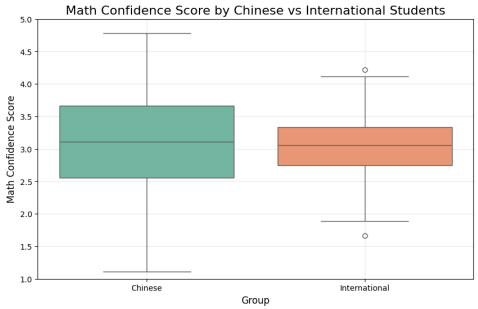
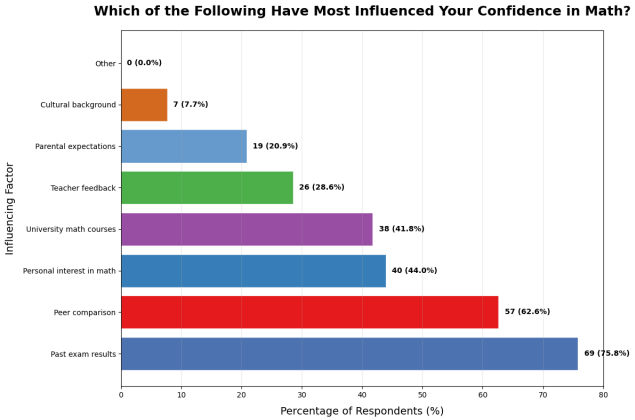


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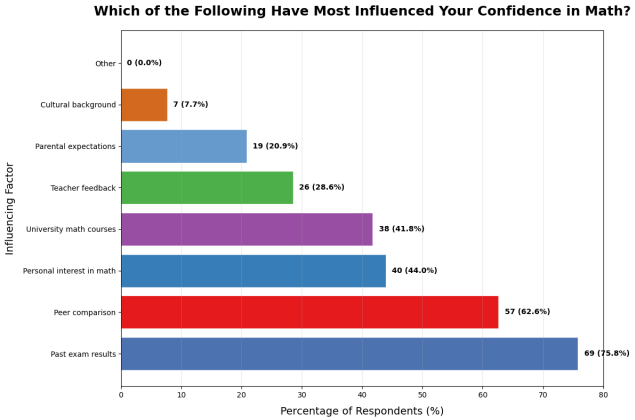
RQ4: What Drives Confidence?



Top Factors:

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Cultural Background?

Only **7.7%** selected it.

Confidence is driven by **immediate academic experiences**, not family/cultural background.

Conclusion: Environment Outweighs Culture

- **Confidence** is a primary driver of major selection in flexible academic environments.
- The lack of difference between Chinese and International students challenges the "East Asian Paradox" in this specific context.
- Confidence is shaped by **tangible academic validation** (grades/exams) and **social comparison**, rather than abstract cultural expectations.

Potential Impact & Implications

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For Educators & Policy Makers

- **Environment:** Reduce the emphasis on public ranking/comparison to protect students' confidence.

For Academic Advisors

- They should encourage students to focus more on their genuine interests and building confidence.

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For Academic Advisors

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Final Thought: By addressing the confidence issue, we can help students make major choices that truly reflect their potential $\Longrightarrow$ **Academic & Career Success**

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Thank You!

Questions?

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